



```
pip install tf-keras
```

```
Requirement already satisfied: tf-keras in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tensorflow<2.21,>=2.20 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tensorboard~2.20.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages
```

```
from sentence_transformers import SentenceTransformer, util
model = SentenceTransformer("all-MiniLM-L6-v2")
```

```
modules.json: 0%|          | 0.00/349 [00:00<?, ?B/s]
```

config_sentence_transformers.json: 0%| | 0.00/116 [00:00<?, ?B/s]

README.md: 0%| | 0.00/10.5k [00:00<?, ?B/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated

sentence_bert_config.json: 0%| | 0.00/53.0 [00:00<?, ?B/s]

config.json: 0%| | 0.00/612 [00:00<?, ?B/s]

model.safetensors: 0%| | 0.00/90.9M [00:00<?, ?B/s]

Loading weights: 0%| | 0/103 [00:00<?, ?it/s]

tokenizer_config.json: 0%| | 0.00/350 [00:00<?, ?B/s]

vocab.txt: 0%| | 0.00/232k [00:00<?, ?B/s]

tokenizer.json: 0%| | 0.00/466k [00:00<?, ?B/s]

special_tokens_map.json: 0%| | 0.00/112 [00:00<?, ?B/s]

config.json: 0%| | 0.00/190 [00:00<?, ?B/s]

paraphrases = util.paraphrase_mining(model,sentences)
paraphrases[:20]

```
[[0.8939038515090942, 3, 6],  
 [0.6787886619567871, 0, 4],  
 [0.6313821077346802, 8, 44],  
 [0.5843963623046875, 13, 45],  
 [0.5794311761856079, 33, 38],  
 [0.5284280180931091, 12, 32],  
 [0.5188640356063843, 29, 38],  
 [0.5187947750091553, 35, 46],  
 [0.5095502138137817, 2, 7],  
 [0.5038483142852783, 8, 50],  
 [0.5013786554336548, 14, 42],  
 [0.48317253589630127, 29, 33],  
 [0.4814086854457855, 15, 30],  
 [0.478863000869751, 22, 28],  
 [0.4751392900943756, 10, 18],  
 [0.4729399085044861, 5, 44],  
 [0.47274249792099, 14, 22],  
 [0.4725485146045685, 28, 34],  
 [0.46822240948677063, 36, 49],  
 [0.4525766968727112, 8, 22]]
```

```
s1 = "The is on the mat"  
s2 = "A cat is sitting on the mat"  
emb1= model.encode(s1)  
emb2 = model.encode(s2)
```

```
sentences = [  
    "The cat sits outside",  
    "A man is playing guitar",  
    "I love pasta",  
    "The new movie is awesome",  
    "The cat plays in the garden",  
    "A woman watches TV",  
    "The new movie is so great",  
    "Do you like pizza?",  
    "She enjoys reading books",  
    "He is walking his dog",  
    "The weather is nice today",  
    "Let's go to the beach",  
    "I need a cup of coffee",  
    "They are planning a trip",  
    "She cooks delicious meals",  
    "I am learning to play the piano",  
    "He works at a tech company",  
    "This book is very interesting",  
    "The sun is shining brightly",  
    "It's raining outside",  
    "The children are playing soccer",  
    "He is fixing the car",
```

"She writes beautiful poetry",
 "The flowers are blooming",
 "I enjoy hiking in the mountains",
 "He is swimming in the pool",
 "We are watching a comedy show",
 "The dog is barking loudly",
 "She is painting a landscape",
 "He likes to play video games",
 "I am studying for my exams",
 "They are building a new house",
 "The coffee shop is busy",
 "He listens to classical music",
 "She is designing a website",
 "I bought a new laptop",
 "They are baking cookies",
 "The bird sings in the morning",
 "He loves going to concerts",
 "She practices yoga every day",
 "The phone is ringing",
 "I need to finish this project",
 "The restaurant has great food",
 "He collects rare coins",
 "She enjoys watching documentaries",
 "They are exploring the city",
 "The laptop is charging",
 "He is training for a marathon",
 "I meditate every morning",
 "The bakery sells fresh bread",
 "She is learning French"

]

```

for i in paraphrases[0:20]:
    score,i,j=i
    print("{}\t\t\t\t {} \t\t\t\t score : {:.4}".format(sentences[i],sentences[j])
  
```

The new movie is awesome	The new movie is so great	
The cat sits outside	The cat plays in the garden	
She enjoys reading books	She enjoys watching documentaries	
They are planning a trip	They are exploring the city	
He listens to classical music	He loves going to concerts	
I need a cup of coffee	The coffee shop is busy	score
He likes to play video games	He loves going to concerts	
I bought a new laptop	The laptop is charging	score
I love pasta	Do you like pizza?	score : 0.5096
She enjoys reading books	She is learning French	
She cooks delicious meals	The restaurant has great food	
He likes to play video games	He listens to classical music	
I am learning to play the piano	I am studying for my exams	
She writes beautiful poetry	She is painting a landscape	

The weather is nice today
A woman watches TV
She cooks delicious meals
She is painting a landscape
They are baking cookies
She enjoys reading books

The sun is shining brightly
She enjoys watching documentaries
She writes beautiful poetry
She is designing a website
The bakery sells fresh bread
She writes beautiful poetry

```
model = SentenceTransformer("distiluse-base-multilingual-cased-v1")
```

```
modules.json: 0%|          | 0.00/341 [00:00<?, ?B/s]
```

```
config_sentence_transformers.json: 0%|          | 0.00/122 [00:00<?, ?B/s]
```

```
README.md: 0%|          | 0.00/2.24k [00:00<?, ?B/s]
```

```
sentence_bert_config.json: 0%|          | 0.00/53.0 [00:00<?, ?B/s]
```

```
config.json: 0%|          | 0.00/556 [00:00<?, ?B/s]
```

```
model.safetensors: 0%|          | 0.00/539M [00:00<?, ?B/s]
```

```
Loading weights: 0%|          | 0/100 [00:00<?, ?it/s]
```

```
tokenizer_config.json: 0%|          | 0.00/452 [00:00<?, ?B/s]
```

```
vocab.txt: 0%|          | 0.00/996k [00:00<?, ?B/s]
```

```
tokenizer.json: 0%|          | 0.00/1.96M [00:00<?, ?B/s]
```

special_tokens_map.json: 0%| | 0.00/112 [00:00<?, ?B/s]

config.json: 0%| | 0.00/190 [00:00<?, ?B/s]

config.json: 0%| | 0.00/114 [00:00<?, ?B/s]

2_Dense/model.safetensors: 0%| | 0.00/1.58M [00:00<?, ?B/s]

```
corpus1 = ["The stock market saw a big drope",
           "Shares declined sharply due to economic news. ",
           "Mony gets money in stock",
           "It may rain today"]
```

```
corpus_emb=model.encode(corpus1)
```

```
# Copute paiwise consine similarity
```

```
cos_similarity = util.pytorch_cos_sim(corpus_emb,corpus_emb)
```

```
avg_sim = cos_similarity.mean(dim=1)
```

```
outlier_index = avg_sim.argmin().item()
```

```
outlier_index
```

```
print(f"outlier sentence:{corpus1[outlier_index]} (AVG Similarity:{avg_sim[ou
```

```
outlier sentence:It may rain today (AVG Similarity:0.33744609355926514)
```

```
corpus1 = ["The dog is outside the home",
           "The Dog is not at home. ",
           "The Dog play outside the Home",
           "The cat playing with baby"]
```

```
corpus_emb=model.encode(corpus1)
```

```
cos_similarity = util.pytorch_cos_sim(corpus_emb,corpus_emb)
```

```
avg_sim = cos_similarity.mean(dim=1)
```

```
outlier_index = avg_sim.argmin().item()
```

```
outlier_index
```

```
print(f"outlier sentence:{corpus1[outlier_index]} (AVG Similarity:{avg_sim[ou
```

```
outlier sentence:The cat playing with baby (AVG Similarity:0.45012956857681274
```

```
corpus1 = ["Ram is a good boy",
           "Ram is very cute boy. ",
           "Ram is playing with childrens",
           "Today clouds are so rainy"]
```

```
corpus_emb=model.encode(corpus1)
```

```
cos_similarity = util.pytorch_cos_sim(corpus_emb, corpus_emb)
avg_sim = cos_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()
outlier_index
print(f"outlier sentence:{corpus1[outlier_index]} (AVG Similarity:{avg_sim[outlier_index]})

outlier sentence:Today clouds are so rainy (AVG Similarity:0.3339882791042328)
```